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TECHNICAL REPORT

Changing the planting date as a climate change adaptation strategy for rice production in Kurunegala district, Sri Lanka

W. R. S. S. Dharmarathna · Srikantha Herath · S. B. Weerakoon

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Abstract The effect of changing the planting date on the dry season rice yield was simulated by using the software Decision Support System for Agrotechnology Transfer (DSSAT 4.5) for four rice varieties grown in Kurunegala district, Sri Lanka under expected climate change. Daily weather data up to the year 2090 were downscaled to the district from Global Climate Model outputs under the emission scenarios A2 and B2 published by the Intergovernmental Panel on Climate Change using the Statistical Downscaling Model (SDSM 4.2). The DSSAT model was applied to simulate future rice yields from four rice varieties grown in the district under three different planting dates: (1) planting in May—the base condition; (2) advancing the planting date by 1 month, i.e., to June; and (3) planting 1 month earlier, i.e., in April. Results show that the seasonally averaged dry season rice yield would increase compared to the base condition when the planting date is advanced by 1 month and, on the other hand, the seasonally averaged rice yield would decrease compared to the base condition when the planting date is delayed by 1 month for all four varieties under both A2 and B2 scenarios. Advancing the rice planting date by 1 month for all four rice varieties can be identified as a non-cost climate change adaptation strategy for rice production in Kurunegala district.

Keywords Rice yield · DSSAT · Planting date · Climate change adaptation · SDSM

Introduction

Rice is the staple food of Sri Lankans, thus it is essential to understand the adverse impacts of climate change on the country's rice production and to find suitable adaptation strategies. Sri Lanka has experienced a temperature increase of 0.016 °C per year in the period between 1961 and 1990 over the entire country according to the Fourth Assessment Report (AR 4) of the Intergovernmental Panel on Climate Change (IPCC) (Cruz et al. 2007). In addition, the expected increase in atmospheric carbon dioxide (CO_2) concentrations under the IPCC emission scenarios and the erratic patterns of rainfall are causing uncertainty regarding the country's rice production.

Temperature rise will impact rice yields negatively in tropical parts of South Asia where rice varieties already being grown close to their temperature tolerance threshold. While direct impacts are associated with rising temperatures, indirect impacts such as water availability, changing soil moisture status and pest and disease incidence are likely to be felt. The most significant impacts are likely to be borne by small-holder rain-fed farmers who constitute the majority of farmers in this region and possess low financial and technical capacity to adapt to climate variability and change (Kelkar and Bhadwal 2007).

Farmers can adapt to climate change to some degree by shifting planting dates, choosing varieties with different growth duration, or changing crop rotations (Wassmann

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and Dobermann 2007). Many researchers have shown that changing the planting date of rice from present practice could be a very good solution to improve rice yield under the impacts of climate change (Desiraju et al. 2010; Mitin 2009; Tripathy et al. 2009; Parry et al. 2004). In Vietnam, Salim (2009) highlights the need to change rice crop sowing dates and management procedures to optimize rice yields under changing climate conditions. In the northern part of Vietnam, the sowing date of spring rice is to be advanced by 15–25 days while the sowing date of summer rice is to be delayed by 20–25 days as the rice growing period expands with changing climate (Salim 2009). By analyzing rice yields at two different sites in the Indian regions of Cuttack and Jorhat under three different climate change scenarios, Krishnan et al. (2007) show that rice yield can be improved when the sowing date of rice is advanced from 1 June to 15 July and 1 July in Cuttack and Jorhat, respectively. Similar studies have shown a decrease in rice yield in Laos, Cambodia and Vietnam but an increase in Thailand when the planting date is advanced under climate change scenario A2. On the other hand, delaying the planting date shows an increase in rice yield in Cambodia and Vietnam but a decrease in Laos and Thailand under climate change scenario A2. Results under climate change scenario B2 are similar to that of A2 except in Thailand where the yield increases when the planting date is delayed and decreases when the planting date is advanced (Mainuddin 2011).

In addition, simulation analysis for developing strategies for adapting rice to climate change scenarios emphasized low-cost adaptation strategies, which included improved crop variety, improved crop management, efficient utilization of irrigation and fertilizer, increased seed replacement by the farmers and increased fertilizer application (Aggarwal et al. 2010). Further studies on increasing the tolerance of spikelet fertility to temperature increase showed rice yield improvements at ten different sites in eastern India (Krishnan et al. 2007).

Rice is produced in Sri Lanka under tropical climate conditions in two seasons per year: the wet (*Maha*) season from October to March, and the dry (*Yala*) season from April to September. The sown extents of major paddy cultivation districts of the country, viz. Kurunegala, Anuradhapura, Polonnaruwa, and Ampara were considered in order to select the study area. Analysis of 30 years worth of data from 1978–1979 wet season to 2008 dry season (i.e., 60 growing seasons), shows that Kurunagala district has the highest sown extent, with around 66,500 ha in wet seasons and around 41,500 ha in dry seasons. Since dry season sown extent is about 37 % less compared to wet season, introducing good adaptation strategies would increase dry season rice production of Kurunegala district. Therefore Kurunegala district was selected for the study. Figure 1 shows the variation of paddy sown extents of major paddy cultivation districts during the aforementioned 60 growing seasons.

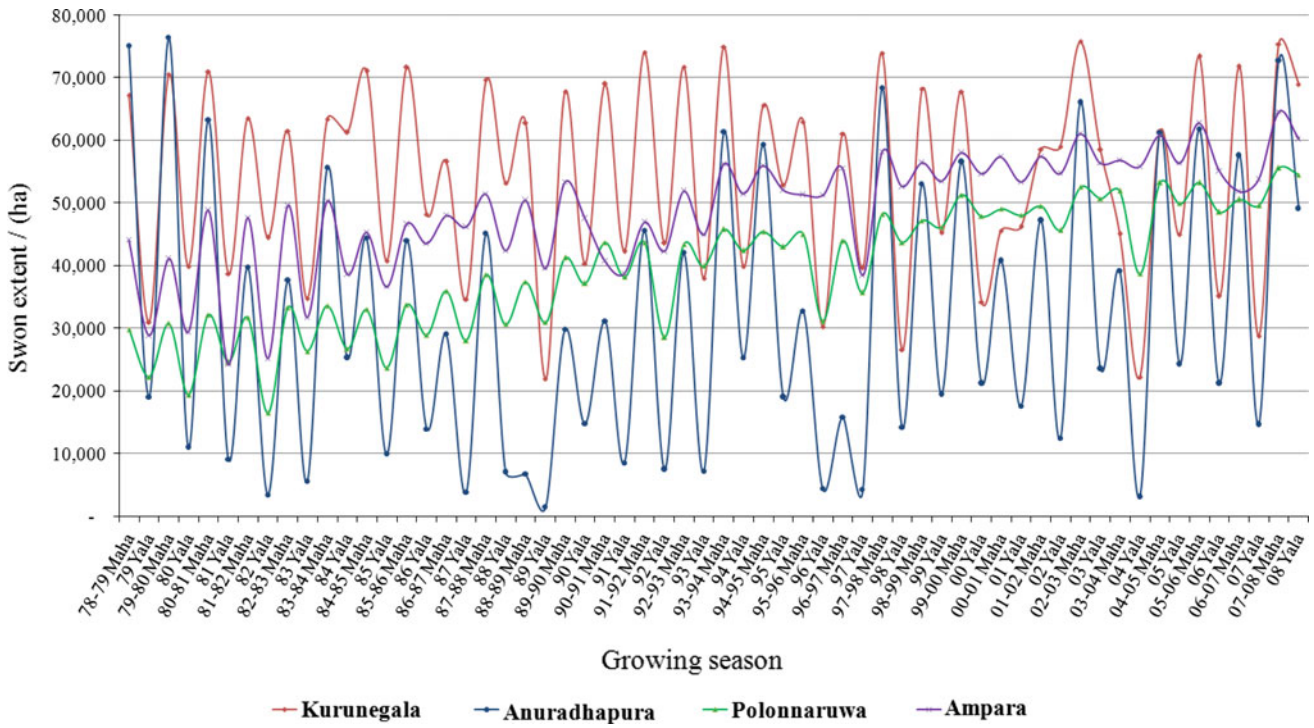


Fig. 1 Variation of paddy sown extents in major paddy cultivation districts. *Source* Department of Census and Statistics (2010)

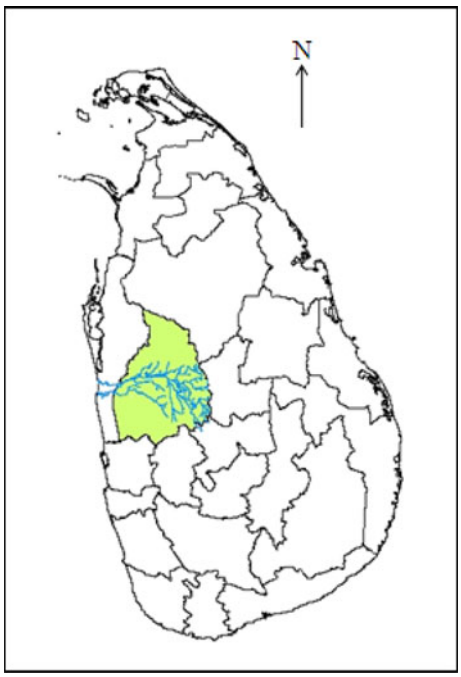


Fig. 2 Location of Kurunegala district and the Deduru oya in the administrative district map of Sri Lanka

Kurunegala district (07.52°N, 80.43°E) is situated in the Northwestern Province in Sri Lanka and has an average daily mean temperature of between 28 and 32 °C with annual rainfall between 1,750 and 2,500 mm. Major irrigation schemes with over 1,000 (,2400) small village tank- and diversion-based storage irrigation systems, fed mainly by Deduru oya is used for paddy cultivation (Sriyananda 2004). The location of Kurunegala district and Deduru oya in the administrative district map of Sri Lanka are shown in Fig. 2.

The decadal mean air temperature of Kurunegala district shows an increasing trend of 0.041–0.065 °C during the period from 1869 to 2007 (De Costa 2008) while precipitation trends are constant with fluctuations in all monsoons (Southwest, Northeast and inter monsoon) during the period of 1860–2000 (Malmgren et al. 2003).

In addition, growth chamber studies carried out in Batalagoda in Kurunegala district, Sri Lanka, have proved that spikelet temperatures above 31 °C decreases pollen fertility and thereby decreases spikelet fertility; the magnitude of decrease depends on the variety. Erratic rainfall and increased evaporative demand lead to increased inland salinity. Moreover, predicted increases in sea levels could result in many paddy lands being affected by coastal salinity. Therefore, the need for salt-tolerant cultivars will increase with climate change (Weerakoon and De Costa 2009).

The aim of this paper was to understand the effect of different planting dates on rice yield under changing climate conditions, and to establish it as a better adaptation strategy for improving rice production while providing a firm background for policy makers to change their cropping calendars.

Methodology

Statistical Downscaling Model (SDSM) 4.2 (Wilby and Dawson 2007) was used for downscaling regional climate and Decision Support System for Agrotechnology Transfer (DSSAT) 4.5 (Hoogenboom 2010) was used for simulating rice growth, under IPCC emission scenarios A2 and B2 and to predict future rice yield under the forecasted weather conditions (Tragoolram et al. 2011). Rice yield variations were analyzed under these conditions for three different planting dates: (1) planting in May—base condition; (2) advancing the planting date by 1 month, i.e., to June; and (3) planting 1 month earlier, i.e., in April.

SDSM 4.2 downscaling model

SDSM 4.2 was used to downscale daily weather variables, viz. maximum temperature, minimum temperature and rainfall, for the Kurunegala district, from the GCM outputs of Hadley Centre experiments (HadCM3) for A2 and B2 scenarios that are available at coarse spatial resolutions for the period up to year 2090. Statistical downscaling consists of step-wise process. The first process is development of statistical relationships between local climate variables (e.g., surface air temperature and precipitation), and large scale predictors (e.g., pressure fields). The second process is application of such relationships to the output of GCM experiments to simulate local climate characteristics in the future (Hoar and Nychka 2008).

The SDSM modeling approach is summarized below (Hessami et al. 2007). The model process can be conditional on the occurrence of an event (i.e., for precipitation) or unconditional (i.e., for temperature). Hence, the modeling of daily precipitation involves two steps: modeling precipitation occurrence followed by modeling of precipitation amounts (if precipitation occurs).

$$O_i = \alpha_0 + \sum_{j=1}^{n} \alpha_j p_{ij} \quad (1)$$

$$R_i^{0.25} = \beta_0 + \sum_{j=1}^{n} \beta_j p_{ij} + e_i \quad (2)$$

where O_i is the daily precipitation occurrence, R_i are daily precipitation amounts, p_{ij} are predictors, n is number of predictors, α and β are model parameters and e_i is modeling error.

The modeling of daily temperature is performed in one step as,

$$T_i = \gamma_0 + \sum_{j=1}^{n} \gamma_j p_{ij} + e_i \quad (3)$$

where T_i is the daily temperature (maximum, minimum or mean) and γ is the model parameter. Once the deterministic component is obtained, the residual term e_i is modeled under the assumption that it follows a Gaussian distribution:

$$e_i = \sqrt{\mathrm{VIF}/_{12}} z_i S_e + b \quad (4)$$

where z_i is a normally distributed random number, S_e is the standard error of estimate, b is the model bias and VIF is the variance inflation factor. The equations used for calculating b and VIF are as follows:

$$b = M_{\mathrm{obs}} - M_d \quad (5)$$

$$\mathrm{VIF} = \frac{12(V_{\mathrm{obs}} - V_d)}{S_e^2} \quad (6)$$

where V_{obs} is the variance of the observations during the calibration period, V_d is the variance of the deterministic part of model output during the calibration period, S_{e} is the standard error, M_{obs} and M_{d} are the mean of the observations and the mean of deterministic part of the model output during the calibration period, respectively (Hessami et al. 2007).

DSSAT 4.5 crop model

The DSSAT modeling system is an advanced physiologically based rice crop growth simulation model that has been applied widely to understand the relationship between rice and its environment. The model is capable of estimating the yield of irrigated, non-irrigated rice and 29 other crops. Further, the model can be used to determine duration of growth stages, dry matter production and portioning, root system dynamics, effect of soil water and soil nitrogen content on photosynthesis, carbon balance and water balance (Basak 2009).

In addition, DSSAT takes in amounts and timing of application of non-labor and non-equipment production factors such as seeds and fertilizer, as well as detailed data on inherent soil quality and climatic conditions. It then employs physical and biophysical models of soil–plant atmosphere interactions to simulate, day by day, the biological growth of the plant by computing crop-specific growth responses, measured precisely under laboratory conditions (Felkner et al. 2009).

Growth duration of a plant is governed by thermal time, and is expressed in growing degree days (GDD).

$$\mathrm{GDD} = \sum_{i=1}^{n} (T_{\mathrm{di}} - T_{\mathrm{base}}) \quad (7)$$

where T_{di} is the daily averaged temperature and T_{base} is the base temperature. The base temperature is considered as 9 °C for rice. The biomass production is calculated within the model as follows.

$$\mathrm{PCARB} = \mathrm{RUE} \times \mathrm{IPAR} \quad (8)$$

$$IPAR = PAR \times \left[1 - e^{(-K*LAI)}\right] \quad (9)$$

$$\mathrm{CARBO} = \mathrm{PCARB}\, x\, \mathrm{Min}(\mathrm{PRFT}, \mathrm{SWDFI}, \mathrm{NDEFI}, 1) \quad (10)$$

where PCARB is the potential biomass production in g m^{-2} day^{-1}, RUE is the radiation use efficiency in g MJ^{-1}, PAR is the photosynthetically active radiation in MJ m^{-2} day^{-1}, IPAR is the fractionPAR intercepted by the plants, K is the extinction coefficient (0.65 for rice), LAI is the green leaf area index, CARBO is the actual biomass production in g m^{-2} day^{-1}, PRFT is the temperature reduction factor (0 or 1), SWDFI is the soil water deficit factor (0 or 1) and NDEFIis the nitrogen deficit factor (0 or 1) where 0 represents the maximum deficit and 1 is for no deficit (Hoogenboom et al. 2003).

Results and discussion

The SDSM and DSSAT models were calibrated and validated using observed data. SDSM was calibrated and validated using daily weather data (maximum temperature, minimum temperature and rainfall) collected from the Department of Meteorology, Colombo for the periods from 1961 to 1980 and 1981 to 2000, respectively. An uncertainty analysis was carried out by examining SDSM predictions for dry seasons; the respective statistical parameters are shown in Table 1.

The statistical parameters, first quartile (Q1), mean and third quartile (Q3) of observed and predicted weather variables lie in close ranges for both calibration and verification periods. Graphical representation of the statistical parameters above is shown as box plots in Fig. 3 for the calibration period.

The DSSAT model was calibrated and validated for four rice varieties using crop and management data collected from the Rice Research and Development Institute (RRDI) farm, Batalagoda for the dry seasons of 2010 and 2006, respectively. The rice varieties Bg 250 (2.5 months), At 307 (3 months), Bg 357 (3.5 months) and Bg 379-2 (4 months) were selected in order to represent both short term and long term rice varieties that are cultivated in Kurunegala district. Figure 4 shows the comparison of

Table 1 Statistical parameters for uncertainty analysis

Parameter	Maximum temperature/(°C)			Minimum temperature/(°C)			Seasonal rainfall/(mm)		
	Observed	A2	B2	Observed	A2	B2	Observed	A2	B2
Q1 (1961–1980)	31.0	31.0	31.0	24.0	24.0	24.0	864.0	935.1	946.3
Mean (1961–1980)	31.4	31.3	31.3	24.0	24.1	24.1	1,024.8	1,029.2	1,043.4
Q3 (1961–1980)	31.8	31.6	31.6	24.3	24.3	24.3	1,077.8	1,137.0	1,083.7
Q1 (1981–2000)	31.4	31.2	31.2	23.9	24.1	24.1	735.3	992.4	989.5
Mean (1981–2000)	31.7	31.7	31.6	24.1	24.2	24.2	986.9	1,062.9	1,056.2
Q3 (1981–2000)	31.9	31.9	31.9	24.3	24.3	24.3	1,144.5	1,116.4	1,147.9

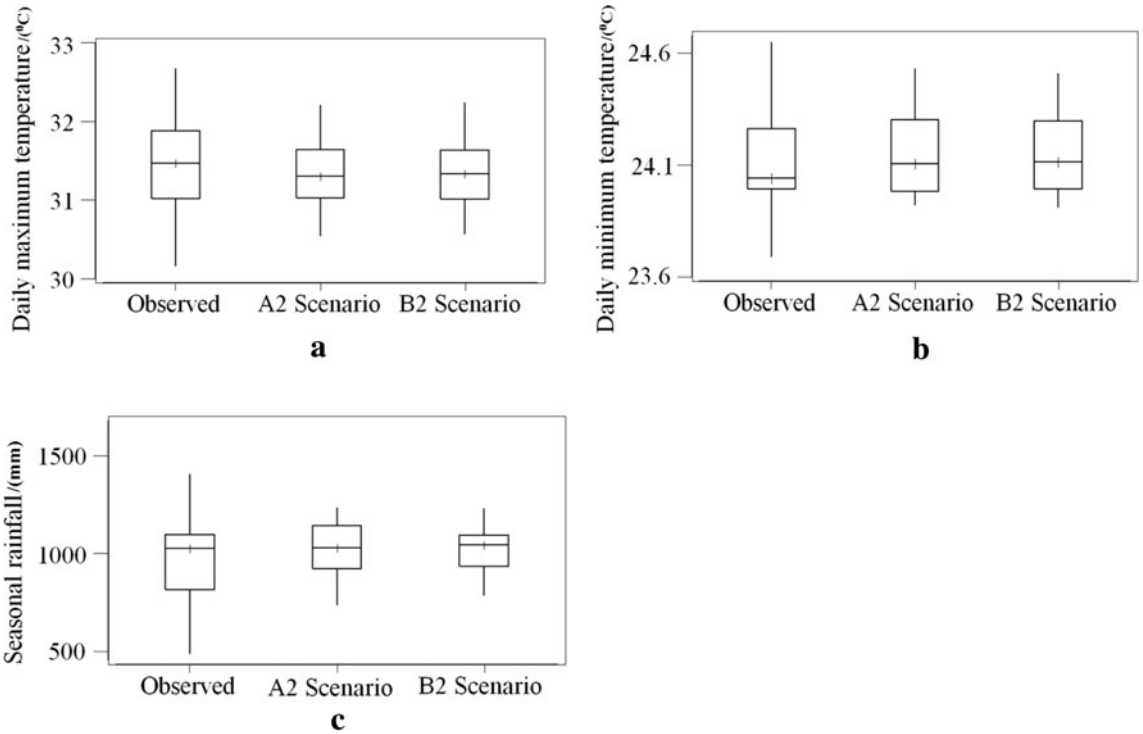


Fig. 3 Graphical representation of statistical parameters for calibration period. **a** Daily maximum temperature. **b** Daily minimum temperature. **c** Seasonal rainfall

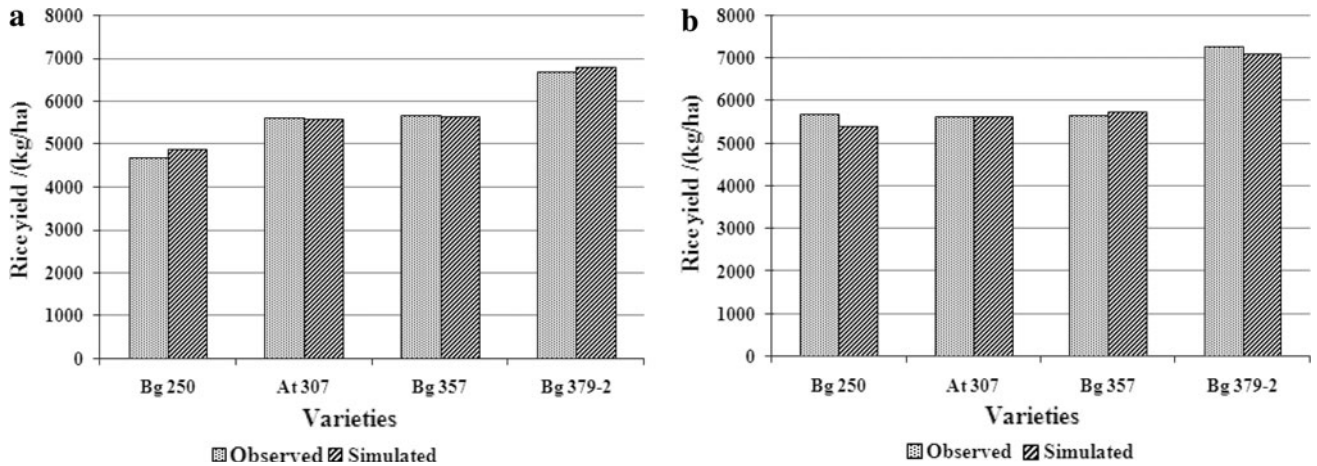


Fig. 4 Comparison of **a** observed and **b** simulated rice yields

observed and simulated rice yield for calibration and validation seasons.

The rice varieties were simulated in DSSAT by using future predicted weather conditions and atmospheric CO_2 concentrations under the emission scenarios A2 and B2 up to 2090. The variations in atmospheric CO_2 concentrations were obtained from the IPCC (2011) report and the respective variations are shown in Fig. 5.

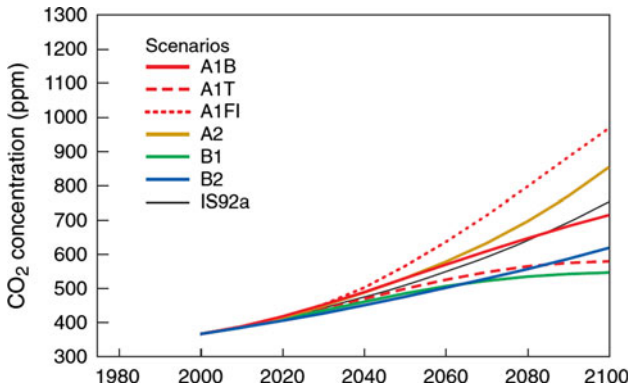


Fig. 5 Atmospheric CO_2 concentrations under Intergovernmental Panel on Climate Change (IPCC) emission scenarios. *Source* IPCC, 2001

The forecasted daily maximum temperature in Fig. 6 shows increasing trends under both A2 and B2 scenarios. The increasing rate of maximum temperature under the high emission scenario A2 has a higher value of 0.05 °C per year compared to that of 0.03 °C per year under the low emission scenario B2.

Forecasted daily minimum temperature in Fig. 7 also shows increasing trends under both A2 and B2 scenarios. The rate of increase of daily minimum temperature is 0.02 °C per year under scenario A2 and is less than the rate of increase in the daily maximum temperature.

The forecasted annual rainfall given in Fig. 8 does not show a significant increasing or decreasing trend, in contrast to minimum and maximum temperatures. The trend under scenario A2 shows a marginally increasing trend while it remains constant under scenario B2 throughout the period.

Simulations were carried out for three cases to examine the effect of planting date on the rice yield: (1) planting date was kept at 26 May as in the observed data and used as the base condition; (2) shifting the planting date forward by 1 month; and (3) shifting the planting date backward by 1 month.

Simulated yields of all four rice varieties under A2 and B2 scenarios for all three cases over the simulation period

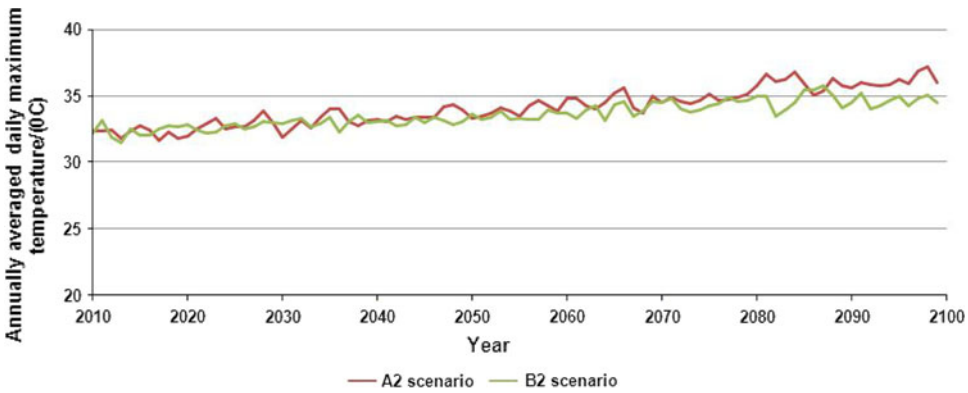


Fig. 6 Forecasted variation of daily maximum temperature under A2 and B2 scenarios

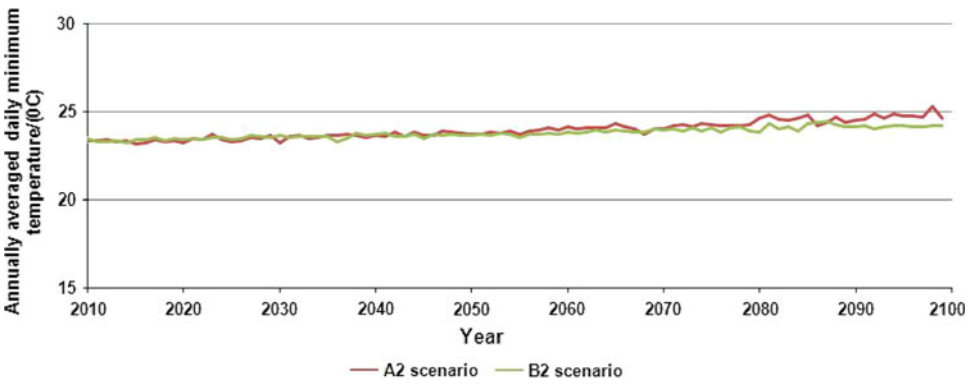


Fig. 7 Forecasted variation of daily minimum temperature under A2 and B2 scenarios

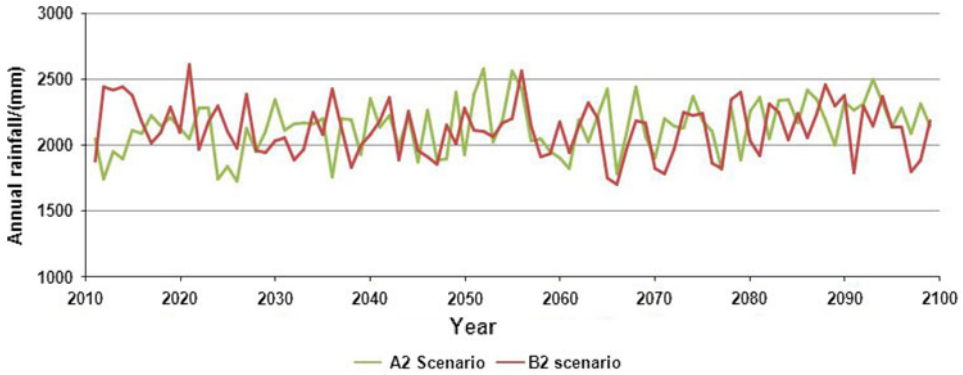


Fig. 8 Forecasted trend of annual rainfall under scenarios A2 and B2

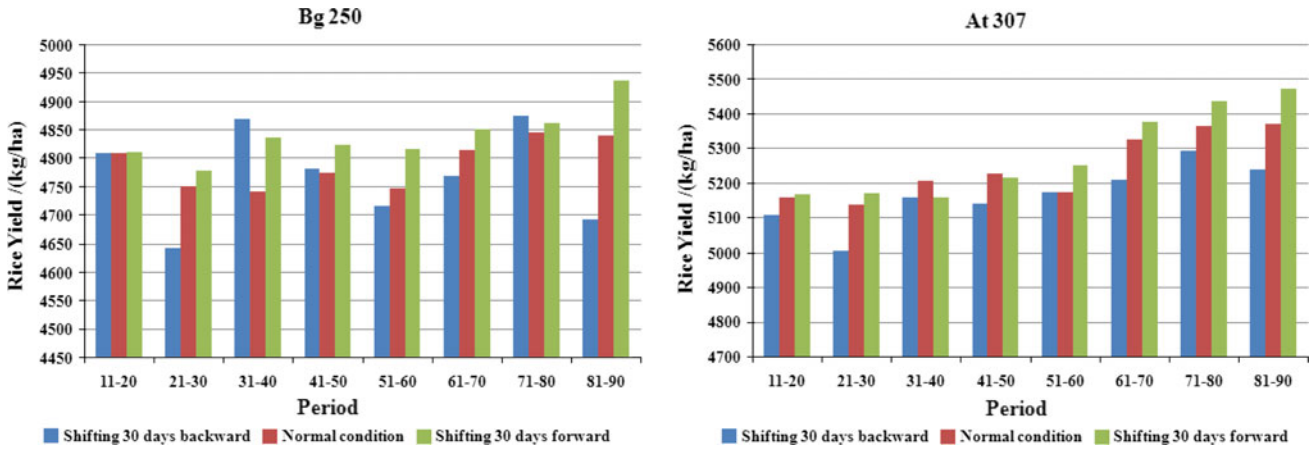


Fig. 9 Rice yield variations for varieties Bg 250 and At 307 under A2 scenario

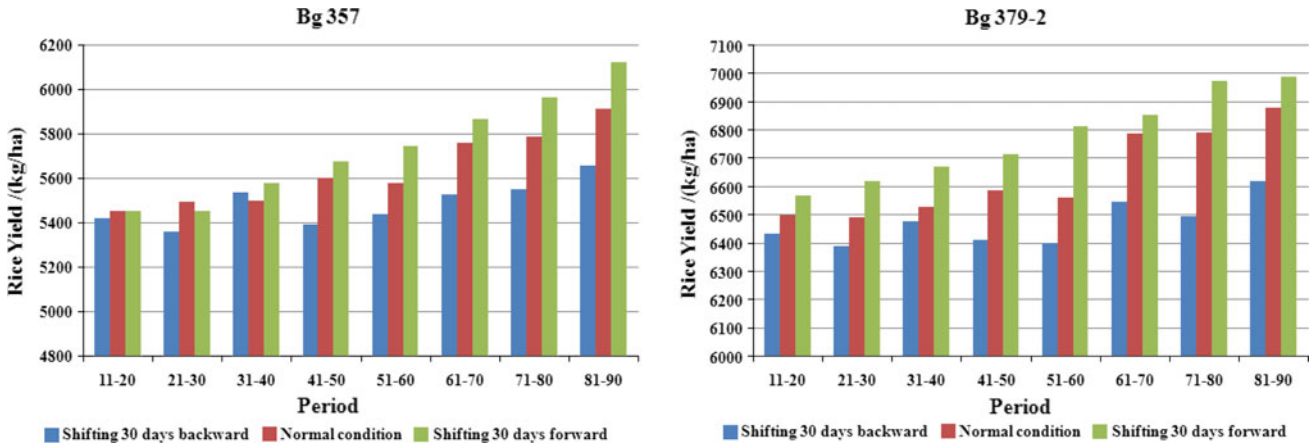


Fig. 10 Rice yield variations for varieties Bg 357 and Bg 379-2 under A2 scenario

from 2011 to 2090 were averaged into 10-year periods and plotted in order to compare the yield variations (Figs. 9, 10, 11, 12).

Figures 9, 10, 11, 12 show the increase in rice yield for Case (2), i.e., shifting the planting date forward by 30 days, in 27 ranges out of 32 ranges, i.e., 84 % of the ranges, under A2 scenario and in 30 ranges out of 32 ranges, i.e., 94 % of the ranges, under B2 scenario. The rice yield decreased when shifting the planting date backward by 30 days in 28 ranges, i.e., 88 % of ranges under scenario A2 and in 29 ranges, i.e., 91 % of ranges under scenario B2.

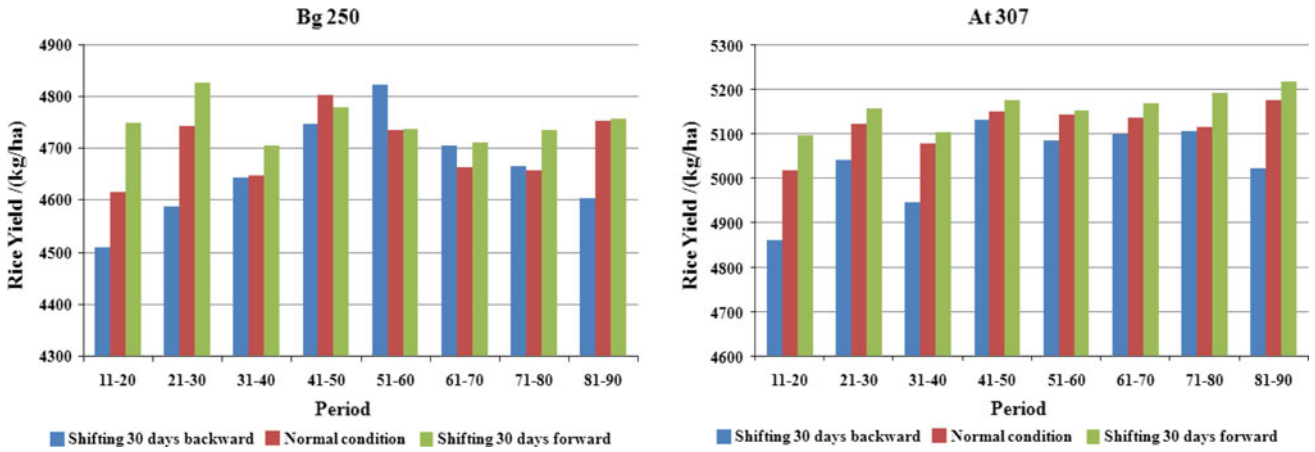


Fig. 11 Rice yield variations for varieties Bg 250 and At 307 under B2 scenario

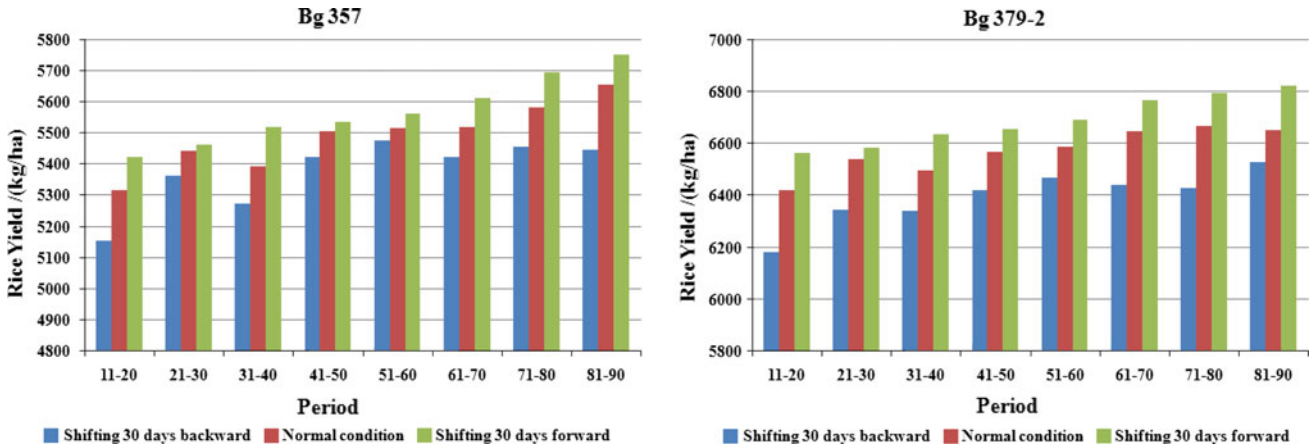


Fig. 12 Rice yield variations for varieties Bg 357 and Bg 379-2 under B2 scenario

Table 2 Deviation of averaged yearly rice yield from the base condition as percentage

Case	Under A2 scenario				Under B2 scenario			
	Bg 250	At 307	Bg 357	Bg 379-2	Bg 250	At 307	Bg 357	Bg 379-2
2	1.0	0.7	1.7	2.0	1.0	0.8	1.4	1.7
3	−0.4	−1.5	−2.7	−2.6	−0.9	−1.6	−2.1	−2.7

The averaged yield variations in case (2) and (3) with respect to case (1) are shown as percentages in Table 2.

The results show that rice yield increased compared to the base condition when the planting date is shifted by 1 month forward for all four varieties under both A2 and B2 scenarios, with a maximum of 2 % (135 kg/ha). Meanwhile, the yield decreased when the planting date is shifted backward by 1 month in all cases, with a maximum decrease of −2.7 % (170 kg/ha). These changes in volumes of rice yield would make a significant impact on the overall production of rice in the district.

Once the dry season from April to September (or *Yala* season) is advanced by 1 month, the wet season from October to March (or *Maha* season) could also be advanced by 1 month so that farmers can harvest the Maha season cultivation in March of the following year, just before the festive month of April.

Conclusions

The effect of changing the planting date as a climate change adaptation strategy for rice production in Kurunegala district, Sri Lanka was investigated under emission scenarios A2 and B2 published by IPCC. Statistical downloading model, SDSM 4.2 was used to downscale the future daily weather data up to 2090. The DSSAT model was used to simulate rice yield from four short-term and long-term rice varieties grown in Kurunegala district.

The results show that seasonally averaged dry season rice yield in the district is:

1. increased by 130 kg/ha when the planting date is shifted 1 month forward
2. decreased by 170 kg/ha when the planting date is shifted 1 month backward

Shifting of the rice planting date forward by 1 month in the two paddy cultivation seasons of the year is recommended as a non-cost climate change adaptation strategy for all four rice varieties in Kurunegala district.

Acknowledgments This study was supported by the Mitsui Fund of UN-CECAR programme of United Nations University—Institute for Sustainability and Peace, Tokyo, Japan.

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